

FAQ on DUT: Frequently Asked Questions about the “Decreasing Universe Theory” (DUT)

João Carlos Holland de Barcellos

1 — Does DUT require the “Jocaxian Nothingness” (JN)?

A: Not necessarily. DUT and JN are independent theories.

However, for historical reasons, DUT emerged — both as a theory and as a physical reality — from JN. That is, DUT was conceived from a study of the consequences of JN.

And although it is a theory independent of JN, we can say that DUT itself (its contraction premises), its physical reality, is explained as having originated through JN.

So, we can say that DUT encompasses JN as the theoretical framework underlying this vision of the Universe.

DUT states that the initial origin of the universe (or multiverse) would be JN.

2 — Does DUT deny the BIG BANG?

A: DUT does not deny the *possible* existence of the Big Bang. *But it denies the Big Bang as being the origin of the Universe.* The true origin of the Universe would be the “Jocaxian Nothingness,” which could, *later on*, cause the Big Bang as a consequence of its “schizo-creations.”

3 — How would we arrive at our current Universe starting from JN?

A: JN generated a multitude of laws and physical elements until these laws and elements stopped(?) being produced. We currently have a certain dynamic equilibrium with some temporal stability. However, from the first “schizo-creations” until we arrived at the physical laws and the current state of the universe — with stars, galaxies and planets — is an area that remains open. How this evolution occurred is still unknown. It may have happened through a (schizo-created) Big Bang, or not.

The schizo-creations may or may not have stopped happening. A law may have schizo-emerged stating, for example: “*The next schizo-creation can*

only occur after X billion years” or “No further schizo-creation can be generated in this Universe,” or so on...

4 — According to DUT, is the Universe expanding, static, or shrinking?

A: What DUT **states** is that the universe is **NOT** expanding **acceleratedly**.

However, despite the gravitational field continuously contracting — very slowly — everything it is present in, this does not mean the universe cannot be expanding: *it may well be expanding*.

With this new theory, the millions of existing measurements will gain new interpretations, and we will be able to determine whether the Universe is expanding or not.

5 — According to DUT, does the gravitational field shrink space itself?

A: Since the gravitational field shrinks EVERYTHING within the space it covers, it is **AS IF** space itself contracted along with it. However, unlike the current model (Λ CDM), which in a way rehabilitated the ETHER — where space stretches and pushes galaxies apart, etc., as if it were something physical — in DUT, space can retain its old form as an unchanging vacuum. In this sense, space itself would not actually be altered by the gravitational field; only the physical elements contained within it would be affected.

That is: in DUT, it is assumed that the gravitational effect acts directly on the physical scales of material systems. Under this interpretation, **it is not necessary** to postulate that space itself contracts.

This is especially true since it would not make much sense: if space itself were shrunk, additional space would have to be created to fill in the space that was shrunk. Therefore, the hypothesis of space itself shrinking is unnecessary.

But people are so used to thinking of space as an ether that stretches and shrinks that, at this point, arguing about it would cause a lot of unnecessary friction. (It's possible that this will become necessary in the future, but I don't believe so.)

6 — Does DUT destroy General Relativity (GR)?

A: If DUT is confirmed, the geometric interpretation of gravity in General Relativity may need to be broadened or modified to incorporate a continuous contraction of physical scales.

This is because, if DUT is true, it introduces a condition that is not — or was not — predicted by GR: *the continuous contraction of “space” by the gravitational field*. In other words, at a **minimum**, GR would be incomplete and would need to be revisited. This opens room for a more complete set of equations to be built.

7 — With the size of the atom and other particles shrinking, what would happen to Quantum Mechanics (QM)? Wouldn't this contraction be incompatible with QM?

A: The answer is that even under the Theory of Relativity, a reference frame that is accelerating causes atoms to *also contract in the direction of motion* — and as far as I know, no one has ever questioned this. Put another way:

The existence of a globally observed change of scale is not, by itself, incompatible with quantum mechanics. Relativity already provides an example in which an observer assigns different dimensions to a complete physical system, without this requiring any modification of the local quantum equations. Therefore, an objection to DUT based on quantum mechanics would also have to apply to the theory of relativity.

8. What is “Comoving Distance” (CD) in the DUT, and what is it used for?

Answer: CD is nothing more than a measurement of distance (relative to the scale of the Earth) at any given fixed time, where that distance is no longer altered by the contraction of the Earth's scales over time.

In the original formula:

$$D = (Z + 1)D_0$$

(where D_0 is the comoving distance), D_0 is the distance measured by the Earth (using the Earth's scale *at that particular time*) at the moment when the photon leaves the star.

In order to compare comoving distances, they would all have to be converted to the same fixed date. Therefore, if we take the fixed date to be, for example, *today's* date on Earth, then we can say that measurements made by the Earth **TODAY** are comoving distances for that exact time, today.

Thus, comoving distances are not particularly useful for calculations unless they are **ALL** converted to a standard reference date (time).

References

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